Technology Stack (Architecture & Stack)

Metadata Attribute	Details
Date	19 October 2026
Team ID	
Project Name	Health360: Clinical & Healthcare Risk Analysis
Maximum Marks	2 Marks

Define Your Technology Stack

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

Technology Stack Specifications

S.No	Component	Description	Technology	Justification / Purpose
1	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python, SQL, HTML5, CSS3, JavaScript	Python and SQL are used for exploratory clinical data analysis, cohort aggregations, and metric validations; HTML5, CSS3, and JavaScript (Chart.js) provide a customized, accessible, responsive client-side interface.
2	Database / Data Warehouse	Warehouse/Store used to store and manage structured	Structured CSV Datastore (`smart_healthcare_datas`	Stores 5,000 comprehensive patient records containing demographic cohorts,

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		business data for analysis and reporting.	`et.csv`) & In-Memory Relational Engine	physiological vitals, and lifestyle risk variables with high portability and rapid query execution.
3	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Data Connector & Native File Ingestion Pipeline	Ingests, checks schemas, and maps 5,000 clinical records seamlessly into Tableau Public desktop and web runtimes with zero data degradation.
4	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Calculated Fields & Python (Pandas) Data Transformation Logic	Formulates clinical categorization (6 Age cohorts: Teenager to Senior), systolic blood pressure thresholds, binary fatigue incidence, and normalized risk index formulas (Mean 100.0).
5	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Public & Chart.js Analytical Engine	Renders 6 core clinical visualization assets (BP Bar Chart, Demographic Pie, Fatigue Stacked Bar, Exercise Donut, Substance Bar, and Lifestyle Risk Heatmap) alongside a 5-point interactive clinical story.
6	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	GitHub Pages (Web Portal) & Tableau Public Cloud Dashboard	Delivers a public, zero-cost, highly responsive cloud deployment allowing clinicians and public health analysts to cross-filter vitals and view clinical findings across any device.